

ACCESSING AND UTILISING GBIF DATA FOR BIODIVERSITY CONSERVATION

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WHAT THIS SESSION COVERS

- **Where the data are** — what GBIF holds for Kosovo, and who put it there.
- **How to get it out** — four routes, and which one to use for which job.
- **How to read it** — what 46,831 records can and cannot tell you.
- **A worked example** — a reproducible site built on exactly this data, for exactly this territory.
- **The traps** — five mistakes that produce a plausible answer and no error message.

The worked example is live at jonasgaigr.github.io/kosovo-biodiversity-data-workshop-2026, and the code that builds it is open.



GETTING THE DATA OUT

One aggregator, four doors, and only one of them gives you a citation.



WHAT GBIF IS

- An **aggregator**, not a survey. Museums, herbaria, monitoring schemes and citizen-science platforms publish; GBIF indexes and serves.
- Every record keeps its **publisher, licence and identifier**. GBIF adds a taxonomic backbone and a quality-flag layer on top.
- For the territory of Kosovo: **46,831 quality-checked records, 4,095 species**, from **125 datasets** and **80 publishers**.

- **eBird alone is 26,009 records** — 56 % of everything, and 219 species.
- 90.7 % of all records are human observations; 1,748 are preserved specimens.
- Kingdoms: 42,003 animals, 3,740 plants, 1,006 fungi.

The national evidence base is, in the main, other people's birdwatching. That is not a criticism of the data — it is the single most important thing to know before interpreting it.



FOUR WAYS IN

Route	Best for	Account	Ceiling
Portal — gbif.org	Exploring, checking a record, one-off extracts	for downloads	none
REST API — api.gbif.org	Live queries, maps, name matching, services	no	100,000 per search
Download service	The full extract, reporting, citation	yes	none
R / Python clients	Scripted, repeatable pipelines	for downloads	as above

The four are the same index seen through different doors. The difference that matters for an institution is that only the download service mints a **DOI**.



THE PORTAL: GBIF.ORG

- Faceted search over **occurrences**, **species**, **datasets** and **publishers** — filter by country, taxon, year, basis of record, licence.
- The **occurrence map** is the same tile service the API serves, so what you see is what you can script.
- **Download** offers Simple CSV, a full Darwin Core Archive, a species list and Parquet.
- Every dataset page carries its **licence**, **publisher** and **DOI**.

Start here, always. Before writing a line of code, look at the map and the dataset list for your area. Ten minutes on the portal tells you whether the data will answer your question at all — and that is cheaper to learn now than after a pipeline has been built.

A registered account is free and is needed only to *download*. Searching, mapping and browsing need nothing.



THE API: NO KEY, NO ACCOUNT

Endpoint	What it gives you
/v1/occurrence/search	Paged records, every portal filter available as a parameter
/v1/species/match	A name resolved against the GBIF backbone — the one call to know
/v2/map/occurrence/...	Ready-made density tiles for a web map
/v1/dataset, /v1/organization	The registry: who published what, under which licence
/v1/occurrence/download/request	Asynchronous extracts, including SQL downloads

- Read access needs **no key and no registration** — it can sit behind a national portal without a credential to manage.
- [/occurrence/search](#) paging stops at **100,000 records**. Past that, the download service is not an optimisation, it is the only option.



THE DOWNLOAD SERVICE, AND WHY THE DOI MATTERS

- You post a **predicate** — taxon, area, year, quality flags — GBIF builds the extract server-side and emails a link.
- The result carries a **permanent DOI**. Citing it credits every contributing institution and lets any reader retrieve the *identical* dataset.
- No size ceiling. In R: `occ_download()` and `occ_download_get()`.

This is the route for anything official. A report whose evidence base cannot be re-fetched is not reproducible, and an assessment that cannot be re-run cannot be defended.

The Kosovo extract used here is **doi.org/10.15468/dl.wzpexq** — 48,607 raw records, built once and reused by every subsequent run.



WHICH ROUTE FOR WHICH JOB

The question	The route
Has anyone recorded <i>Lynx lynx</i> near this site?	Portal, or /occurrence/search
Draw a live density map in our own web portal	/v2/map tiles
Is this name current, and what is its key?	/species/match
Give me every record in the country, citably	Download service + DOI
Rebuild the same analysis next year	rgbif / pygbif in a script
Who published the records we rely on?	/dataset , /organization
Which listed invasive species are recorded just across the border?	/occurrence/search with facets

One rule of thumb. If the answer will end up in a document that someone else has to trust, take the route that gives you a DOI.



READING THE RECORDS HONESTLY

Occurrence data answer a narrower question than they appear to.



PRESENCE, NOT ABSENCE

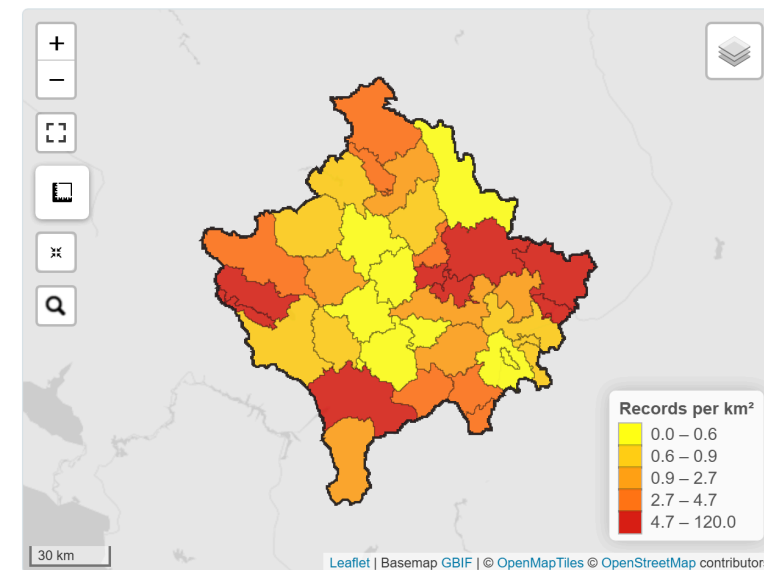
- A record is a statement that **someone found this taxon here, then, and published it**. Nothing more.
- An empty map cell means one of three things, and the data cannot tell you which: nobody looked; someone looked and found nothing; someone found something and never published it.
- **Absence of records is not evidence of absence**. For screening a development proposal, that distinction is the whole ball game.
- Records also cannot carry abundance unless the publisher recorded it — and most did not.



EFFORT, NOT RICHNESS

- **Graçanicë / Gračanica:** 14,672 records, 120 per km² — the densest municipality in the country.
- **Mamushë / Mamuša:** 1 record.
- 13,346 of Graçanicë's records come from **one wetland of 1.10 km²**.
- 81 % of dated records are from 2015 or later; trend analysis before then rests on thin ground.

Dense cells mark well-visited places, not rich ones. Read the density map as a map of where people go.



HOW PRECISELY ARE THE RECORDS PLACED?

Stated coordinate uncertainty	Records	Per cent	Good enough for
100 m or better	3,180	6.8	a site boundary, once verified
100 m – 1 km	2,193	4.7	screening at 1–5 km
1 – 10 km	9,364	20.0	screening at 10 km
coarser than 10 km	90	0.2	presence in the country
not stated	32,004	68.3	screening, after a check

- Two thirds of records **say nothing about their own accuracy**. That is not the same as being accurate.
- By precision alone, **one record in fifteen** could help draw a Natura 2000 boundary — and **one Annex I bird record in twenty**. GBIF is a screening layer, and a good one.



FITNESS FOR USE: SCREEN, DON'T JUST SELECT

- The download was selected on the **GADM** boundary for Kosovo, because that is the filter GBIF offers. GADM's outline is a generalisation.
- Screened against the **OpenStreetMap** border — 60 m median from Eurostat's independent digitisation, against GADM's 587 m — **1,776 records (3.7 %)** fall outside the country.
- They are not borderline: **94 % carry a publisher-assigned country code of ME, MK, RS or AL**, a median 552 m beyond the border.
- The opposite error cannot be repaired: **412 km² of Kosovo lies outside GADM's polygon**, so records there were never in the download.

Select with the tool's filter. Screen with the best boundary you have. They are not the same polygon, and using one for both hides both kinds of error.



THE SITE

A reproducible evidence base for one territory, built from the routes above.



WHAT IT IS

- A **static website**: summary statistics, seven interactive maps, a browser-side record explorer and open data downloads.
- **One R pipeline** — acquire, screen, annotate, export, credit — and **one Quarto document**. Re-running does not mint a new DOI.
- **No server, no database, no hosting cost** — and nothing to keep patched.
- Audience: ministry and agency staff who need the evidence, not the code.

Biodiversity of Kosovo

An analysis of GBIF occurrence data in support of conservation planning

[Code](#)

On this page

Why this matters

- Data provenance and citation
- Summary statistics
- Data quality control
- Where the records are
- Survey coverage by municipality
- Protected areas
- Species protected under the EU Nature Directives
- Extinction risk: the IUCN Red List
- Explore the records
- The species checklist
- Who published these records
- Download the data
- Methods and reproducibility

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Why this matters

Kosovo sits within the Balkan biodiversity hotspot, one of the most species-rich regions in Europe. Evidence-based conservation — designating protected areas, screening development proposals, and reporting against the EU Nature Directives — depends on knowing **which species occur where**.

The Global Biodiversity Information Facility (GBIF) aggregates species occurrence records published by museums, research institutions, monitoring programmes and citizen-science platforms worldwide. This report brings the GBIF holdings for the territory of Kosovo into a single, quality-controlled, openly downloadable evidence base, and highlights the records that relate to species protected under the EU Birds and Habitats Directives or assessed as threatened on the IUCN Red List.

46,831

QUALITY-CHECKED
RECORDS

4,096

SPECIES RECORDED

5,073

DIRECTIVE-LISTED
RECORDS

37

IUCN-THREATENED
SPECIES

38 / 38

MUNICIPALITIES WITH
RECORDS

1826–2026

TEMPORAL COVERAGE

Data provenance and citation

The occurrence data were obtained through the GBIF asynchronous download service, which mints a permanent

SEVEN THEMATIC EXTRACTS

Extract	Records	Species
Overall biodiversity	46,831	4,095
Birds (class Aves)	28,785	255
Birds Directive, Annex I	4,215	70
Habitats Directive (all annexes)	858	78
Habitats Directive, Annex II	395	38
IUCN threatened (CR, EN, VU)	650	37
EU invasive alien species (Union list)	24	5

Each is published as **GeoPackage, CSV and Excel**, and each carries the municipality, protected area, Red List category, annex membership and Union-list status as columns — so the subset can be used without re-running any of the matching.

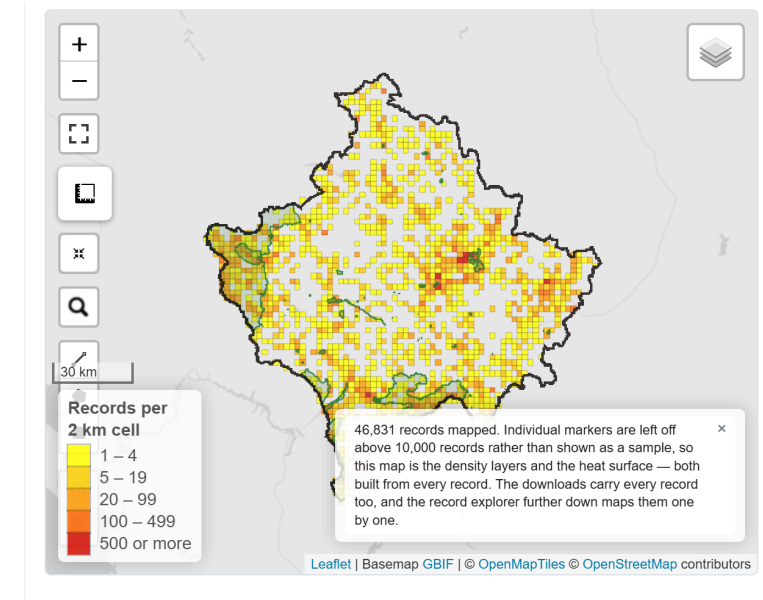


MAPS THAT ANSWER DIFFERENT QUESTIONS

Five switchable layers on every map:

- **Occurrence records** — pop-up with determination, place, stated accuracy and a link through to GBIF
- **Record density** — 2 km grid, classed
- **Heat surface** — the shape of sampled and unsampled ground
- **Protected areas** — a wash under the records
- **Municipalities** — with counts on hover

Basemaps, colours and the density ramp are GBIF's own, taken from its published brand steps and its live tile service.



PROTECTED AREAS: WHERE THE EVIDENCE IS



- The register is the **European Environment Agency's inventory of nationally designated areas** — Kosovo's own official return.
- **20 of the 48 mapped sites carry any occurrence record at all.**
- 63 % of everything inside the network comes from one 1.10 km² wetland. Set it aside and the concentration is real but modest.

SPECIES OF COMMUNITY INTEREST

- The annex lists are **generated from the consolidated EUR-Lex texts**, not hand-maintained: Birds I (193 taxa), Birds II (83), Habitats II (887), IV (370), V (76).
- Every name resolved against the GBIF backbone, with the asterisked priority species and the listing rank preserved.

Why not the ready-made checklists on GBIF? They are geographically filtered. The Birds Annex I checklist holds 128 of the 193 listed taxa — and the omissions are exactly the ones that matter here. *Alectoris graeca* is absent, as are the Balkan endemics. Using it would have quietly under-reported the most conservation-relevant taxa in the country.



BIRDS: FROM RECORDS TO AN SPA CASE

What the records give

- **4,215** Annex I bird records of **70** species — **82 %** from eBird, which states **no uncertainty** at all.
- The richest site, **Ligatina e Hencit**, holds **43** of the 70 species, and is **not an IBA**.
- Kosovo's IBAs — Sara mountain and Prokletije, with Mokra Gora and Kopaonik on the boundary — are **mountain sites**, estimated in breeding pairs, several from 2008–2013.

What an SPA case needs

- **One bird table**, the layers as views of it: Annex I, regular migrant, SDF population type (**p r c w**), breeding code, a count **with its unit**.
- For each trigger species, a **population estimate from standardised counts** over two or three seasons, and the **criterion** it meets.
- A boundary around **what the birds use** — not the IBA outline.
- **eBird's own dataset** for effort and non-detections; the GBIF copy is presence-only.

INVASIVE ALIEN SPECIES OF UNION CONCERN

- The **Union list** under Regulation (EU) 1143/2014: **114 species** that may not be imported, kept, sold or released. 111 apply today, and three from August 2027.
- In the Kosovo extract: **5 of them, 24 records**. The 12 since 2015 are all iNaturalist or Pl@ntNet; the other 12 come from older fish and mammal datasets — 1983, 1992 or undated.
- *Ailanthus altissima*, tree of heaven: **8 records here, 406 across the four neighbours**. That is a recording gap, not a clean country.

The watch list: recorded next door, not here — 20 species

- **All four neighbours:** paper mulberry, pond sliders
- **Three:** coypu, eastern mosquitofish
- **Serbia, North Macedonia:** common milkweed, black bullhead, Egyptian goose
- **Serbia, Montenegro:** Japanese, giant and Bohemian knotweed
- **Montenegro:** small Asian mongoose, 97 records
- **Albania:** Asian hornet, 2025

The Regulation is built on early detection: only a species caught early can still be eradicated. The counts are live API facets, dated in the report — a screening aid, not a citation.



A COMPLETE CHECKLIST, AND STILL CHECK THE NAMES

- INBO publishes the Union list on GBIF — **114 of 114**, CCO, doi.org/10.15468/97aucj. Unlike the Annex I checklist, it is complete.
- The site's list is built from the **legal text** and agrees with it species for species. The checklist supplies the English names.
- **One accent:** “*Triadica sebífera*” defeats GBIF's name parser. Canonical name *Triadica spec.*, no backbone link.

The American mink is listed as *Neogale vison*, the genus it moved to in 2021. The backbone does not know the combination. The matcher returns the genus *Neogale* — a synonym of *Mustela* — at **94 % confidence, and no error.**

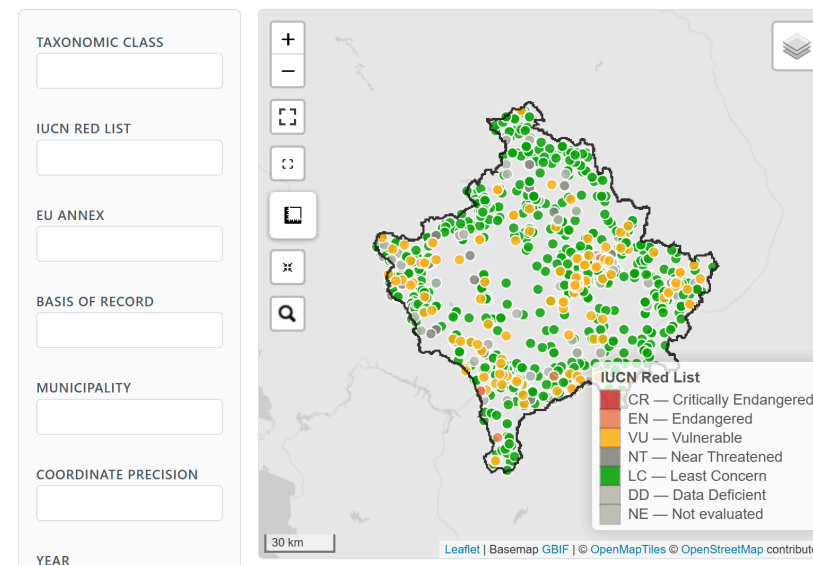
Taken at face value, every native weasel, stoat and polecat becomes an invasive species. Treat **HIGHERRANK** as a failure, never as a match. And EUR-Lex now answers scripts with HTTP 202 and an empty body; the same acts come from Cellar, publications.europa.eu, by content negotiation.



THE RECORD EXPLORER

- Filter by **class, Red List category, EU annex, EU invasive species, basis of record, municipality, coordinate precision and year.**
- Map and table are **linked** — filter one, the other follows.
- All of it runs in the reader's browser. **No Shiny server, nothing to host.**

The idea is borrowed from the GBIF Viewer built for Ukraine by the Habitat Foundation and the Ukrainian Nature Conservation Group — a Shiny application — and re-implemented client-side so it costs nothing to keep running.



OPEN DATA, IN FORMATS PEOPLE ACTUALLY USE

Overall biodiversity 46,831 records · 4,095 species GeoPackage (26.6 MB) CSV (21.7 MB)	Birds (class Aves) 28,785 records · 255 species GeoPackage (16.4 MB) CSV (13.3 MB)
Birds Directive, Annex I 4,215 records · 70 species GeoPackage (2.6 MB) CSV (2.0 MB) Excel (640 kB)	Habitats Directive (all annexes) 858 records · 78 species GeoPackage (600 kB) CSV (427 kB) Excel (144 kB)
Habitats Directive, Annex II 395 records · 38 species GeoPackage (328 kB) CSV (195 kB) Excel (71 kB)	IUCN threatened species (CR, EN, VU) 650 records · 37 species GeoPackage (472 kB) CSV (314 kB) Excel (105 kB)

GeoPackage for GIS, CSV for anything, Excel where the subset is small enough to open. Every record keeps its [license](#) column, and the report credits all 125 datasets and 80 publishers by name.

TRAPS THAT FAIL SILENTLY

Every one of these returns a plausible answer and no error message.



THREE ABOUT KOSOVO SPECIFICALLY

- 1. The GADM code is `XKO`, not `XKX`.** `XKX` is the World Bank / ISO alpha-3 style code. `pred("gadm", "XKX")` returns **zero records — no error, just an empty dataset**. Measured: `gadmGid=XKO` returned 52,290 records, `country=XK` 48,947, `gadmGid=XKX` nothing.
- 2. `CoordinateCleaner` cannot see Kosovo by default.** Natural Earth records Kosovo with `iso_a3 == "-99"` — no assigned code. The country test therefore flags **100 % of Kosovo records**, and with `value = "clean"` returns an **empty data frame with no warning**. Supply your own reference polygon.
- 3. GADM's Kosovo municipalities are the pre-2010 set — 30, not 38.** The single coordinate carrying more records than any other in the country, over thirteen thousand of them, sits on ground that moved to the new municipality of Graçanicë in 2010. Reported through GADM, every one of them landed in Fushë Kosovë — and every coverage map and table said so.



TWO ABOUT METHOD AND TOOLING

4. Never screen against the polygon you selected on. Screening the download against GADM level 0 — the polygon the download was built from — flagged nothing at all. That looked like a clean result and was really the test grading its own paper: every record is inside that polygon *by construction*, so asking whether it is can only return yes.

5. Basemap tiles fail without failing. CARTO's Positron tiles still return HTTP 200 and a valid PNG, but now carry "API KEY REQUIRED" stamped diagonally across every one served to an unauthenticated client. Nothing appears in the console; the map simply looks wrong.

GBIF's own tiles need no key — but they are **512 pixels**, so they need `tileSize = 512` and `zoomOffset = -1`. Leave Leaflet's 256-pixel default and every tile still lands in the right place, at half its designed size, which reads as a rendering fault rather than a configuration one.



WHAT THIS COULD MEAN HERE

Two things to take away, one of them larger than the other.



THE SMALLER ONE: REUSE THE PIPELINE

- The repository is a **template**. Change the area predicate and the species lists and it runs for any territory.
- Everything it depends on is **free and open**: GBIF, OpenStreetMap, the EEA inventory, EUR-Lex, R, Quarto, GitHub Pages. No licence, no server, no procurement.
- It produces, from one command: a quality-controlled evidence base, a citable DOI, seven thematic extracts in three formats, and a public report.
- The **pale areas on the density map are a fieldwork plan** — the cheapest survey prioritisation available, and it comes free with the data you already have.



THE LARGER ONE: PUBLISH, DON'T ONLY CONSUME

- 56 % of Kosovo's national record base is **eBird**. The country's own monitoring holdings are largely absent from the global index.
- Publishing through an **IPT** would put national survey and monitoring data into the same evidence base — visible to reporting, to research and to screening, with the publishing institution credited on every record.
- It is also the cheapest route to **Natura 2000 readiness**: the annex screening shown here runs on whatever is published, and today that is mostly not yours.
- Consuming GBIF answers *what is already known*. Publishing to it is how what you know becomes part of the record.



TAKEAWAYS

- **Take the route that gives you a DOI** for anything that ends up in a document someone else must trust.
- **Select with the tool's filter; screen with the best boundary you have.** They are not the same polygon.
- **Read density as effort, not richness** — and read a blank cell as a question, not an answer.
- **Automate the whole path** from download to published report, so next year's version costs one command.
- **Publishing is the part that changes the evidence base.** Everything else reads it.



THANK YOU FOR YOUR ATTENTION!

jonasgaigr.github.io/kosovo-biodiversity-data-workshop-2026



NATURE IS OUR HERITAGE
AND THE FUTURE

aopk.gov.cz

